Do Now · retrieval

Network types, architecture and hardware

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does an OS device driver do?

Q2

What does a router connect?

Q3

Define a client-server relationship.

Q4

Binary search on sorted indexes 0–14 for 13: show midpoint index sequence.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A LAN covers a local area such as a school building; a WAN connects networks across wider geography.

Core idea 2

Trace — Client-server uses dedicated service systems; peer-to-peer allows peers to exchange roles and resources, with different management implications.

Core idea 3

Explain — NICs connect devices, switches forward local Ethernet frames, wireless APs bridge Wi-Fi, and routers forward IP packets between networks.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LAN — Local Area Network.

WAN — Wide Area Network.

switch — Forwards frames inside local network.

router — Forwards packets between IP networks.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace school laptop reaching an external site through a wired school network.

Follow the worked method

1. Predict — Switch moves frames within LAN
2. Check the answer — router handles packets destined for another network.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

APs provide wireless access

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define LAN, WAN, switch and router. [4]

Question 2

Describe a laptop's network path via wireless access point, switch and router to the Internet. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare client-server and peer-to-peer for school file sharing, giving strengths and limits. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A network switch and router are always interchangeable because they both move packets.

Use key vocabulary from memory

1. Explain LAN in your own words.

2. Explain WAN in your own words.

3. Explain switch in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.